

SYED MAOZ ARIF

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SUMMARY

Hardware Developer and Electronics Engineer experienced in multi-layer PCB design, embedded systems, and Internet of Things (IoT) architecture. Developed 12+ hardware projects and earned 17 technical certifications. Secured First Position in national Complex Engineering Project (CEP) exhibitions. Manage 50+ members serving as Institution of Engineering and Technology (IET) Chapter President, and deliver freelance hardware designs maintaining a 100% on-time delivery rate.

EDUCATION

COMSATS University Islamabad, Abbottabad Campus

B.S. Electrical Engineering (Electronics)

Expected Aug 2028

Abbottabad, Pakistan

- **Honors:** Altium Global Scholarship Awardee (Selected among top 1% globally).
- **Relevant Coursework:** Embedded Systems, Digital Logic Design, Signal Integrity, C/C++ Programming.

EXPERIENCE

President

Institution of Engineering and Technology (IET) On Campus

May 2025 – Present

Abbottabad, Pakistan

- Direct chapter operations and manage a 5-person cross-functional executive team, organizing initiatives for 50+ active members.
- Organize 5+ technical workshops (including Artificial Intelligence in UAVs) with 100+ attendees, increasing student engagement by 40%.

Freelance Hardware & Mechanical Designer

Fiverr & Upwork (Freelance Client Platforms)

Jan 2024 – Present

Remote

- Deliver over 15 multi-layer PCBs and 3D CAD models to global clients, achieving 100% on-time delivery and 5-star ratings across four sectors.
- Optimize component selection (BOM) and schematic captures, reducing overall hardware prototyping costs by 25% for 10+ clients.

PROJECTS

Industrial Edge AI Condition Monitoring

Academic Project | Dual-MCU, GPIO, C/C++

May 2026 – Jul 2026

Abbottabad, Pakistan

- Designed an industrial development board integrating 70 GPIOs, reducing 3-phase induction motor diagnostic time by 40%.
- Secured First Position at Complex Engineering Project (CEP) Exhibition and 2nd Position at EMCOT out of 50+ competing projects.

Stratum-1 Precision Hardware Clock

Independent Project | FPGA, Verilog

Jul 2026 – Aug 2026

Abbottabad, Pakistan

- Engineered a hardware time synchronizer using pure digital logic, bypassing OS latency to achieve 0 ms delay and 100% synchronization accuracy.
- Developed a custom Verilog UART receiver and NMEA state machine on a Lattice FPGA to process live GPS signals directly in silicon.

ElectroParts Inventory Architecture

Independent Project | Firebase, CapacitorJS

Jul 2026 – Aug 2026

Abbottabad, Pakistan

- Architected a cross-platform inventory management system supporting 32 distinct electronics categories, increasing component retrieval efficiency by 50%.
- Implemented dual-storage local/cloud synchronization and linear barcode generation, enabling zero-latency data access for 1000+ logged components.

IncliSense Precision Digital Level

Independent Project | I2C, Arduino, Telemetry

Mar 2026 – Apr 2026

Abbottabad, Pakistan

- Built a precision leveling device analyzing dual-axis spatial data via Inter-Integrated Circuit (I2C), yielding high-accuracy PC telemetry.
- Awarded Global Runner-Up in the Instructables Sensors Contest out of 500+ international submissions.

SKILLS

Hardware Design: Altium Designer, KiCad, Proteus, Schematic Capture, Signal Integrity, PCB Design

Embedded Systems: ESP32, STM32, Arduino, Lattice FPGA, I2C, SPI, UART, MQTT, Firebase, IoT

Programming: C/C++, Python, Verilog (HDL), MATLAB, JavaScript, HTML5/CSS3

3D CAD & Tools: SolidWorks, Fusion 360, AutoCAD, 3D Printing, Git, LaTeX

Soft Skills & Management: Cross-functional Leadership, Stakeholder Communication, Project Management, Agile Methodologies

CERTIFICATIONS

Altium Certified: Designer Essentials <i>Altium Education</i>	Mar 2026 <i>Online</i>
Generative AI Mastermind <i>Outskill</i>	Jan 2026 <i>Online</i>
Arduino & IoT Certification <i>Cardiff Institute</i>	Aug 2025 <i>Online</i>
3D Printing & CAD Engineering Certificate <i>Online Institute</i>	May 2025 <i>Online</i>